

# Executive Summary

## Aviation Emissions & Fuel Efficiency Assessment

### Project Overview

An aviation operations, fuel consumption, and emissions dataset containing 15,121 flight records was reviewed to assess data quality, fuel efficiency, route-level emissions performance, and sustainability reporting readiness.

Prior to analysis, the dataset underwent data quality validation and preparation to ensure reliable reporting of operational, fuel, and emissions metrics.

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### Key Findings

#### 1. Data Quality Issues Required Remediation Before Analysis

Several material data quality issues were identified, including duplicate flight records, invalid distance values, and incomplete fuel consumption data. These issues had the potential to distort fuel-efficiency calculations, emissions totals, and route-level reporting.

A total of 55 records were removed due to duplicate or invalid distance values, while records with missing fuel information were retained and documented as analytical limitations.

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#### 2. Fuel Consumption and Emissions Were Significant Across the Reporting Period

Total fuel consumption across the reporting period was approximately **159.4 million kilograms**, resulting in an estimated **503.6 million kilograms of CO<sub>2</sub> emissions** using standard aviation fuel emission conversion factors.

This highlights the importance of accurate fuel reporting and emissions monitoring for sustainability and regulatory reporting purposes.

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#### 3. Average Passenger Load Factor Was Strong

The fleet achieved an average passenger load factor of approximately **78.8%**, indicating generally effective utilization of available passenger capacity.

Higher load factors improve operational efficiency by distributing emissions across a greater number of passengers and reducing emissions intensity per passenger.

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## 4. Medium-Haul Operations Generated the Highest Share of Emissions

Analysis by distance category showed that **medium-haul flights contributed the largest share of total CO<sub>2</sub> emissions** within the reporting period.

This suggests that medium-haul operations may provide the greatest opportunity for targeted fuel-efficiency initiatives and emissions reduction programs.

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## 5. Emissions Were Concentrated Across Specific Routes and Aircraft Types

The **A320 aircraft type accounted for the highest share of total fuel consumption**, while the route **APT\_119 - APT\_001 generated the highest CO<sub>2</sub> emissions** within the reporting period.

These findings highlight specific operational areas where fuel-efficiency improvements and emissions-reduction initiatives could deliver the greatest impact.

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# Recommendations

## 1. Improve Fuel Data Capture and Completeness

A significant number of records contained missing fuel consumption values. Strengthening data collection processes and validation controls would improve the reliability of fuel-efficiency and emissions reporting.

## 2. Prioritize High-Emission Routes and Aircraft Types

Operational reviews should focus on the highest fuel-consuming aircraft types and the highest-emitting routes to identify opportunities for efficiency improvements and emissions reductions.

## 3. Implement Automated Data Validation Controls

Automated validation checks should be introduced to identify duplicate records, invalid distance values, and other data quality issues before sustainability reporting activities are performed.

## 4. Strengthen Sustainability Reporting Processes

Additional attention should be given to ETS/CORSIA-related reporting fields to improve reporting completeness, transparency, and regulatory compliance readiness.

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## Additional Data Requested

To improve reporting accuracy and sustainability analysis, the following data would be requested from the client:

- Complete fuel burn data for flights with missing fuel records.
  - Aircraft age and engine type information to support efficiency comparisons.
  - Detailed load factor and cargo utilization data.
  - Sustainable Aviation Fuel (SAF) usage information, if applicable.
  - Clarification of ETS/CORSIA reporting fields with high levels of missing data.
  - Route profitability and operational delay data to assess operational efficiency alongside emissions performance.
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## Conclusion

Following data quality remediation, the dataset provided meaningful insight into fuel consumption, emissions performance, and operational efficiency. The findings highlight opportunities to improve fuel reporting quality, strengthen sustainability reporting, and prioritize emissions-reduction initiatives on high-impact routes and aircraft types.